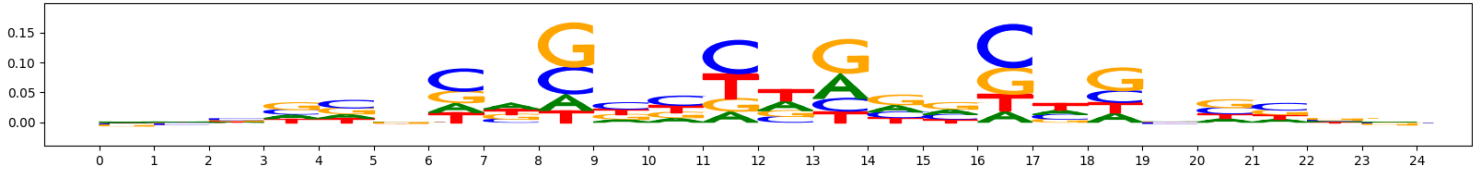


Bias model training and quality check report

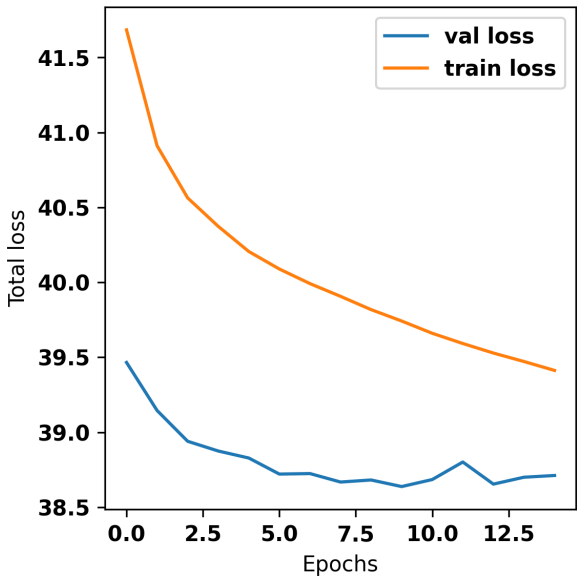
Preprocessing report

The image below should look closely like a Tn5 or DNase bias enzyme motif.



Training report

The val loss (validation loss) will decrease and saturate after a few epochs.



Bias model performance in peaks and non-peaks

Counts Metrics: The pearsonr in non-peaks should be greater than 0 (higher the better). The pearsonr in peaks should be greater than -0.3 (otherwise the bias model could potentially be capturing AT bias). MSE (Mean Squared Error) will be high in peaks.

Profile Metrics: Median JSD (Jensen Shannon Divergence between observed and predicted) lower the better. Median norm JSD is median of the min-max normalized JSD where min JSD is the worst case JSD i.e JSD of observed with uniform profile and max JSD is the best case JSD i.e 0. Median norm JSD is higher the better. Both JSD and median norm JSD are sensitive to read-depth. Higher read-depth results in better metrics.

What to do if your pearsonr in peaks is less than -0.3? In the range of -0.3 to -0.5 please be wary of your chrombpnet_wo_bias.h5 (that wil potentially be trained with this bias model) TFModisco showing lots of GC rich motifs (> 3 in the top-10). If this is not the case you can continue using the chrombpnet_wo_bias.h5. If you end up seeing a lot of GC rich motifs it is likely that bias model has learnt a different GC distribution than your GC-content in peaks. You might benefit from increasing the bias_threshold_factor argument input to the *chrombpnet bias pipeline* or *chrombpnet bias train* command used in training the bias model and retrain a new bias model. For more intuition about this argument refer to the [FAQ](#) section in wiki. If the value is less than -0.5 the [chrombpnet training](#) will automatically throw an error.

counts_metrics		nonpeaks.pearsonr		nonpeaks.mse		peaks.pearsonr		peaks.mse	
		0.12		5.4		-0.58		17.0	
profile_metrics		nonpeaks.median_jsd		nonpeaks.median_norm_jsd		peaks.median_jsd		peaks.median_norm_jsd	
		0.63		0.16		0.49		0.26	

TFModisco motifs learnt from bias model (bias.h5) model

TFModisco motifs generated from profile contribution scores of the bias model. cwm_fwd, cwm_rev are the forward and reverse complemented consolidated motifs from contribution scores in subset of random peaks. These CWM motifs should be free from any Transcription Factor (TF) motifs and should contain either only bias motifs or random repeats. For each of these motifs, we use TOMTOM to find the top-3 closest matches (match_0, match_1, match_2) from a database consisting of both MEME TF motifs and heterogenous enzyme bias motifs that we have repeatedly seen in our datasets. The qvals (qval0,qval1,qval2) should be high (> 0.0001) if the closest hit is a TF motif (i.e indicating that the closest match is not the correct match) - this is also generally verifiable by eye as the closest match will look nothing like the CWMs. The qvals should be low if the closest hit is enzyme bias motif and generally verifiable that the top match looks like the CWM. The first 3-5 motifs in the list below should look like enzyme bias motif.

What to do if you find an obvious TF motif in the list?

Do not use this bias model as it will regress the contribution of the TF motifs (along with bias motifs) from the chrombpnet_nobias.h5. Reduce the bias_threshold_factor argument input to the *chrombpnet bias pipeline* or *chrombpnet bias train* command used in training the bias model and retrain a new bias model. For more intuition about this argument refer to the [FAQ](#) section in wiki.

What to do if you are unsure if a given CWM motif is resembling the match_0 logo for example?

Get marginal footprint on the match_0 motif logo (using the command *chrombpnet footprints* and make sure that the bias models footprint is closer to that of controls with no motif inserted - for examples look at [FAQ](#))

pattern	NumSeqs	cwm_fwd	cwm_rev	match0	qval0	match0_logo	match1	qval1	match1_logo	match2	qval2	match2_logo
pos__0	10723			TN5_2	4.634450e-10		TN5_1	1.937860e-09		TN5_7	0.000835	
pos__1	4488			TN5_2	6.228750e-05		TN5_1	8.340380e-05		TN5_3	0.007771	
pos__2	4240			TN5_8	4.469970e-02		TN5_2	1.163120e-01		TN5_1	0.178643	
pos__3	2246			TN5_3	1.813400e-01		SP5_MOUSE.H11MO.0.C	1.813400e-01		KLF4_MA0039.3	0.181340	
pos__4	1934			TN5_3	7.226950e-07		TN5_1	3.155180e-05		TN5_7	0.194495	
pos__5	798			TN5_3	7.826020e-08		TN5_4	2.109230e-05		TN5_5	0.000021	
pos__6	703			TN5_3	5.564670e-08		TN5_2	1.090690e-03		TN5_1	0.003027	
pos__7	539			BRAC_MOUSE.H11MO.0.B	1.000000e+00		NaN	NaN		NaN	NaN	
pos__8	444			ZN281_MOUSE.H11MO.0.A	2.935880e-04		SP1_HUMAN.H11MO.0.A	1.684280e-03		KLF5_HUMAN.H11MO.0.A	0.001684	
pos__9	371			TN5_3	1.505140e-08		TN5_1	1.171780e-02		TBX21_TBX_3	0.019935	
pos__10	366			RREB1_MA0073.1	2.959630e-01		EGR2_HUMAN.H11MO.0.A	2.959630e-01		EGR1_HUMAN.H11MO.0.A	0.399182	
pos__11	317			PRDM6_HUMAN.H11MO.0.C	5.638470e-02		ZNF384_MA1125.1	5.638470e-02		ANDR_HUMAN.H11MO.0.A	0.114034	
pos__12	274			EGR2_HUMAN.H11MO.0.A	5.217120e-03		ZN281_MOUSE.H11MO.0.A	5.217120e-03		KLF5_HUMAN.H11MO.0.A	0.005217	
pos__13	248			RREB1_MA0073.1	1.000000e+00		BRAC_MOUSE.H11MO.0.B	1.000000e+00		NaN	NaN	
pos__14	235			TN5_3	1.613130e-06		TN5_1	2.915340e-04		TN5_4	0.001389	

pattern	NumSeqs	cwm_fwd	cwm_rev	match0	qval0	match0_logo	match1	qval1	match1_logo	match2	qval2	match2_logo
pos__15	123			Hoxc10.mouse_homeodomain_2	1.272910e-01		HOXC10_homeodomain_2	1.272910e-01		HOXC12_homeodomain_1	0.127291	
pos__16	102			RREB1_MA0073.1	1.000000e+00		BRAC_MOUSE.H11MO.0.B	1.000000e+00		NaN	NaN	
pos__17	41			TN5_7	3.258130e-02		TN5_2	2.850580e-01		HNFB4G_HUMAN.H11MO.0.B	0.808558	
pos__18	30			TN5_4	1.121710e-05		TN5_5	1.121710e-05		TN5_2	0.000319	
pos__19	20			Tcf12_MA0521.1	9.223220e-01		Myog_MA0500.1	9.223220e-01		TFE2_HUMAN.H11MO.0.A	0.922322	

TFModisco motifs generated from counts contribution scores of the bias model. cwm_fwd, cwm_rev are the forward and reverse complemented consolidated motifs from contribution scores in subset of random peaks. These motifs should be free from any Transcription Factor (TF) motifs and should contain motifs either weakly related to bias motifs or random repeats. For each of these motifs, we use TOMTOM to find the top-3 closest matches (match_0, match_1, match_2) from a database consisting of both MEME TF motifs and heterogenous enzyme bias motifs that we have repeatedly seen in our datasets. The qvals should be high (> 0.0001) if the closest hit is a TF motif (i.e indicating that the closest match is not the correct match, this is also generally verifiable by eye and making sure the closest match looks nothing like the CWMs).

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pattern	NumSeqs	cwm_fwd	cwm_rev	match0	qval0	match0_logo	match1	qval1	match1_logo	match2	qval2	match2_logo
pos__0	4645			DNASE_2	4.932820e-01		EVI1_HUMAN.H11MO.0.B	4.932820e-01		EVI1_MOUSE.H11MO.0.B	4.932820e-01	
pos__1	3702			DNASE_2	6.727500e-01		FOXB1_forkhead_2	8.900610e-01		LIN54_MA0619.1	8.900610e-01	
pos__2	3434			ZNF384_MA1125.1	3.381280e-02		PRDM6_HUMAN.H11MO.0.C	8.184310e-02		FOXJ3_HUMAN.H11MO.0.A	8.184310e-02	
pos__3	2519			ZNF384_MA1125.1	7.422320e-02		FOXD2_forkhead_1	5.396050e-01		SOX7_HMG_2	5.396050e-01	
pos__4	2500			FOXB1_forkhead_2	1.072800e-01		DNASE_2	3.726110e-01		FOXB1_MA0845.1	3.726110e-01	
pos__5	2435			ZNF384_MA1125.1	6.154200e-03		DNASE_2	4.917050e-02		DNASE_6	8.915940e-01	